

Question	Answer	Marks
6(a)	- force per unit length - force per unit current - length / current perpendicular to field 1 mark for any two points, 2 marks for all three points	B2
6(b)(i)	into the page	B1
6(b)(ii)	$F = Bqv$	C1
	$= 4.8 \times 10^{-3} \times 1.6 \times 10^{-19} \times 1.7 \times 10^7 = 1.3 \times 10^{-14} \text{ N}$	A1
6(b)(iii)	arrow at point X pointing down the page	B1
6(b)(iv)	$F = mv^2 / r$	C1
	$1.3 \times 10^{-14} = (9.11 \times 10^{-31}) \times (1.7 \times 10^7)^2 / r$	C1
	$(r = 0.020 \text{ m})$ $d = 2r$ $d = 0.040 \text{ m}$	A1
6(c)	path shows upwards deflection such that the curvature is always anticlockwise within the field	B1
	circular path with larger radius	B1
	line enters field at X and leaves field at distance $2d$ vertically from X	B1

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7(c)(i)	D 1.25	A1